



Norwegian
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Project Louis

Articulated Robot Arm



Funded by Eik Lab (pretty please)

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1 Motivation

Some tasks heavily rely on human intuition and coordination, making them difficult to fully automate. However, these tasks can also be dangerous, and require high precision. While the task can't be fully automated, robotics can provide mitigation of risk factors such as exposure to dangerous chemicals, radiation, electricity and moving parts. The goal of this project is to produce a teleoperated robot arm, protecting humans from hazardous environments.

2 Final vision

Our approach for this project is to create a robot arm mimicking a human hand. It will be remote controlled by haptic gloves akin to lucid gloves, and provide a feedback loop that ensures finger resistance matching the robots grip. This will provide high levels of intuition when dealing with problems outlined in our motivation. For rotational tracking we intend to utilize an IMU. For positional tracking we intend to follow the evolving standards of the VR industry.

3 First iteration / Proof of concept

We will create an articulated robot arm, with the ability to operate in the task space of $\mathbb{R}^3$, focusing on kinematics and remote control. It will be controlled using a simple game controller (xbox, playstation, etc). Featuring methods like forward and inverse kinematics the robot will have multiple control modes. These will be an angular control system and a positional system in our task space related to the robot base coordinate system.

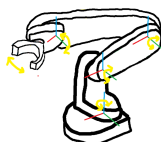


Figure 1: Truly artistic sketch handpainted by Johannes with trackpad and MS Paint

The articulated arm will consist of rigid bodies and rotational joints, following TRR anatomy. Using a transrotational joint at the base, followed by two rotational joints for our arm. The end effector will be equipped with a simple claw attached to a rotational joint.

For the first iteration we will use servo motors as they have the benefit of setting angles with high precision. The servo motors angle limitations will somewhat affect our task space, however it will still be able to operate in task space in $\mathbb{R}^3$.

4 Requested starting capital

We already got our hands on the electronic components, which makes up most of the projects cost (BOM in github). We request an additional 1000kr to cover 3D printing costs.



5 Project links

<https://github.com/JohannesStandal/Eik---Project-Louis>